

CLASS XII · LECTURE 3

Decisions and the switch Statement

Option I · Two periods · Section C asks for switch with a logical diagram

By the end of this lecture you can

- Write if, if-else and else-if for a mark sheet or a larger-of-two test.
- Draw switch as a choice among cases, each closed by break.
- Explain what happens when break is omitted.

if, else, and a chain of tests

A decision runs one block and skips another. The condition in brackets is tested. If it is true (not zero), the first block runs. **else** is the path when the test fails. A chain of **else if** is used when there are several bands, such as grades. The first true test wins, and the rest are skipped.

```
#include <stdio.h>

int main(void) {
    int marks;
    printf("Enter marks: ");
    scanf("%d", &marks);
    if (marks >= 40)
        printf("Pass\n");
    else
        printf("Fail\n");
    return 0;
}
```

Program 3.1 Pass and fail. One test, two paths.

```
#include <stdio.h>

int main(void) {
    int marks;
    printf("Enter marks: ");
    scanf("%d", &marks);
    if (marks >= 80)
        printf("Grade A\n");
    else if (marks >= 60)
        printf("Grade B\n");
    else if (marks >= 40)
        printf("Grade C\n");
    else
        printf("Fail\n");
    return 0;
}
```

Program 3.2 Three grades. The tests are in order, highest band first.

EXAM TIP Use == when you compare

A single = stores a value. A double == compares. **if (choice = 1)** assigns 1 and is always true. **if (choice == 1)** asks the question you meant. This is the most common logic error in a decision.

switch

switch chooses among constant values of one integer or character expression. Each **case** labels one value. **break** leaves the switch. **default** runs when no case matches. It is the right tool when you are matching a menu number or a day number, not when you are testing ranges such as “80 and above”.

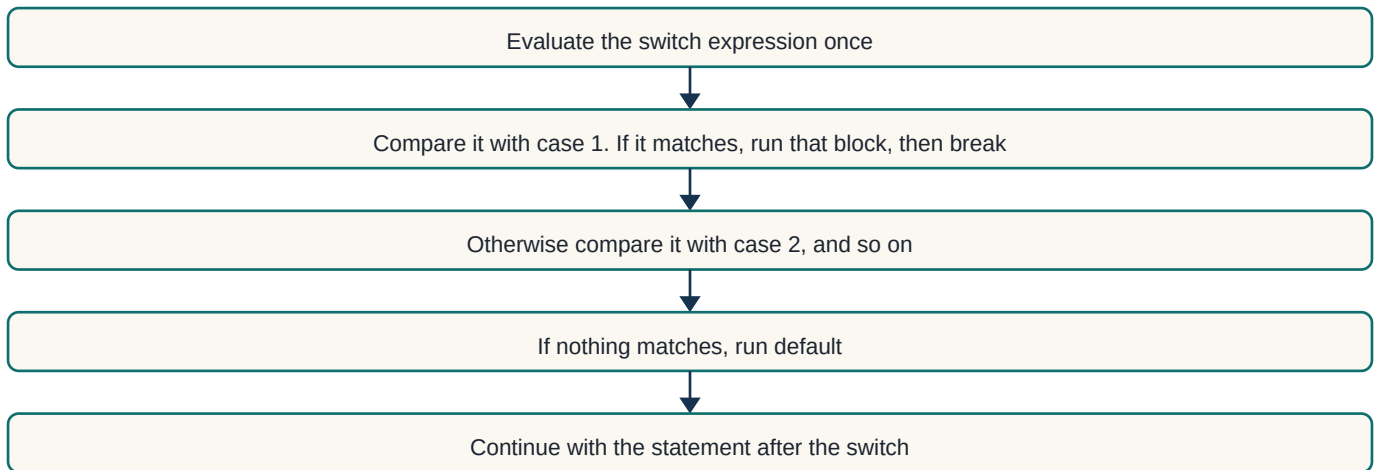


Figure 3.1 Logical diagram of switch. Each case ends in break so control does not fall into the next case.

```

#include <stdio.h>

int main(void) {
    int choice;
    printf("1 Add 2 Subtract: ");
    scanf("%d", &choice);
    switch (choice) {
        case 1:
            printf("You chose add\n");
            break;
        case 2:
            printf("You chose subtract\n");
            break;
        default:
            printf("No such option\n");
    }
    return 0;
}
  
```

Program 3.3 A menu. break stops a match from falling into the next printf.

If you delete the `break` after case 1, a choice of 1 prints both lines. C does not leave the switch at the end of a case by itself. That fall-through is the fact a long answer must mention. `default` does not need a break if it is the last label, but writing one does no harm.

ON THE BOARD 10-mark shape for switch

Define switch in two lines. Draw the diagram: expression, then arrows to case 1, case 2 and default, each case arrow continuing out only after break. Then write Program 3.3. End with one sentence: without break, execution falls into the next case.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Which operator compares two values for equality?

- A) =
- B) ==
- C) !=
- D) +=

Short answers

S1. Why is break written at the end of a case?

Detailed answer

L1. Define the switch statement and draw its logical diagram.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. A single equals sign assigns. Two equals signs compare.

S1. break leaves the switch as soon as that case has run. Without it, C continues into the next case and runs that code as well. This is called fall-through. default handles every value that matched no case.

L1. switch selects one path according to the value of an integer or character expression. The expression is evaluated once and compared with each case label. The matching case runs until break, which leaves the switch. If no case matches, default runs. Diagram: a box for the expression, arrows to case 1, case 2 and default, and from each case an arrow labelled break that joins the statement after the switch. Example: `switch(choice) { case 1: printf("Add"); break; case 2: printf("Subtract"); break; default: printf("Error"); }`. Omitting break makes case 1 fall into case 2.